

# Email Analysis Report

## A Hope to CoCanDa — Phishing Email Investigation

|                |                                                                      |
|----------------|----------------------------------------------------------------------|
| Author         | Mohamed Sikkanthar Asaf                                              |
| Date           | March 2026                                                           |
| Case           | A Hope to CoCanDa — BTLO Challenge                                   |
| Email File     | A Hope to CoCanDa.eml                                                |
| Classification | TLP: WHITE                                                           |
| Verdict        | <span>🔴</span> <b>MALICIOUS — Phishing Email with Spoofed Sender</b> |

### 1. Executive Summary

This report documents the analysis of a suspicious email with the subject "A Hope to CoCanDa" sent to themajornearth@gmail.com. The email was analysed as part of a Blue Team Labs Online (BTLO) challenge to identify phishing indicators and attacker infrastructure.

The email exhibits multiple high-confidence phishing indicators including SPF authentication failure, a mismatched Reply-To address, use of a known malicious mail relay (emkei.cz), and Base64-encoded content containing a suspicious attachment. The sender spoofed the identity of "Bill" using the domain microapple.com — a typosquat likely designed to impersonate Apple Inc.

### 2. Email Header Analysis

The email headers were extracted from the raw .eml file and examined field-by-field. The following table summarises key header fields and their analysis findings.

| Header Field | Value                            | Status     | Analyst Note                                                                               |
|--------------|----------------------------------|------------|--------------------------------------------------------------------------------------------|
| Delivered-To | themajornearth@gmail.com         | INFO       | Intended target recipient                                                                  |
| From         | "Bill" <billjobs@microapple.com> | SUSPICIOUS | Display name 'Bill' used to impersonate. Domain microapple.com is a typosquat of apple.com |
| Reply-To     | negeja3921@pashter.com           | SUSPICIOUS | Differs from From address. Attacker controls this inbox —                                  |

| Header Field              | Value                                 | Status     | Analyst Note                                                                                            |
|---------------------------|---------------------------------------|------------|---------------------------------------------------------------------------------------------------------|
| Return-Path               |                                       |            | replies go to attacker not sender                                                                       |
|                           | <billjobs@microapple.com>             | INFO       | If delivery fails, bounce sent here. Matches From field in this case                                    |
| SPF                       | FAIL                                  | FAIL       | 93.99.104.210 not authorised to send on behalf of microapple.com — strong spoofing indicator            |
| DKIM                      | Not present                           | WARNING    | No DKIM signature found — email integrity cannot be verified                                            |
| DMARC                     | Not evaluated                         | WARNING    | No DMARC policy enforcement observed                                                                    |
| Received (relay)          | emkei.cz [93.99.104.210]              | MALICIOUS  | emkei.cz is a publicly known fake/anonymous email sender service used by threat actors                  |
| X-Priority                | 3 (Normal)                            | INFO       | No urgency manipulation used in headers                                                                 |
| Date                      | Tue, 26 Jan 2021 01:41:18 -0500       | WARNING    | Date field can be spoofed by sender                                                                     |
| Message-ID                | <20210126064118.1993E221F8@localhost> | SUSPICIOUS | localhost in Message-ID confirms email was sent from emkei.cz fake mailer, not a legitimate mail server |
| Content-Type              | multipart/mixed                       | INFO       | Indicates multiple content parts — email contains both text body and attachment(s)                      |
| Content-Transfer-Encoding | base64                                | INFO       | Email body encoded in Base64 — decoded to reveal actual content                                         |

## 3. Email Authentication Analysis

### 3.1 SPF (Sender Policy Framework)

SPF check returned FAIL. The sending IP address 93.99.104.210 (emkei.cz) is not listed as an authorised sender in the SPF record for microapple.com. This is a definitive indicator that the email did not originate from a legitimate microapple.com mail server and the sender identity is spoofed.

- Sending IP: 93.99.104.210
- SPF Result: FAIL
- Expected sender domain: microapple.com
- Conclusion: IP not in authorised SPF record — spoofing confirmed

### 3.2 DKIM (DomainKeys Identified Mail)

No DKIM signature was present in the email headers. DKIM provides cryptographic verification that the email content has not been tampered with in transit. The absence of DKIM means there is no way to verify the integrity of the message body or that it was authorised by the domain owner.

### 3.3 DMARC

DMARC (Domain-based Message Authentication Reporting and Conformance) was not evaluated or enforced in this case. Without DMARC policy enforcement, emails failing SPF and DKIM checks may still reach recipient inboxes depending on the mail provider's configuration.

## 4. Sender Infrastructure Analysis

### 4.1 Sending Server — emkei.cz

The Received header reveals the email was relayed through emkei.cz, a publicly accessible online anonymous email sending service. This platform allows anyone to send emails with any From address without authentication, making it a frequently abused tool by phishing actors and social engineers.

- Server: emkei.cz
- IP Address: 93.99.104.210
- Nature: Anonymous/fake email relay service — known malicious infrastructure
- Threat Actor Usage: Widely used in low-sophistication phishing and BEC attempts
- Recommended Action: Block 93.99.104.210 and emkei.cz at email gateway

### 4.2 Sender Domain — microapple.com

The sender domain microapple.com is a typosquat domain designed to visually resemble apple.com. This is a classic social engineering technique where a threat actor registers a look-alike domain to deceive victims into believing the communication is from a legitimate, trusted organisation.

- Registered domain: microapple.com
- Target impersonation: Apple Inc. (apple.com)
- Technique: Typo squatting / domain spoofing
- SPF: Not properly configured — permitted attacker to spoof

### 4.3 Reply-To Address — negeja3921@pashter.com

The Reply-To address negeja3921@pashter.com differs significantly from the From address billjobs@microapple.com. This is a deliberate attacker technique: when a victim replies to the email, the response is routed to the attacker-controlled inbox rather than back to the spoofed sender. The pashter.com domain is a disposable email service commonly abused by threat actors.

- From: billjobs@microapple.com
- Reply-To: negeja3921@pashter.com
- Discrepancy: HIGH — strong phishing indicator
- Purpose: Capture victim replies without exposing real attacker infrastructure

## 5. Email Content & Attachment Analysis

### 5.1 Body Content — Base64 Decoding

The email body was encoded using Base64 (Content-Transfer-Encoding: base64). Using CyberChef with the recipe From Base64 → To Hex, the encoded content was decoded. The decoded output revealed the plain text body of the email beginning with:

"Hi TheMajorOnEarth, The abducted CoCanDians a..."

This social engineering lure references fictional colony abduction, crafted to create urgency and curiosity in the target recipient.

### 5.2 Attachment Analysis — File Signature (Magic Bytes)

The email contained attachments extracted from the PuzzleToCoCanDa directory. File type was verified using magic bytes (file signatures) rather than relying on file extensions alone, as extensions can be easily spoofed.

| File Name           | Magic Bytes | True File Type | Notes                                           |
|---------------------|-------------|----------------|-------------------------------------------------|
| DaughtersCrown.jpeg | FF D8 FF    | JPEG Image     | Confirmed JPEG. ExifTool metadata extracted.    |
| GoodJobMajor.pdf    | 25 50 44 46 | PDF File       | Magic bytes 25 50 44 46 = %PDF — confirmed PDF. |

### 5.3 ExifTool Metadata — DaughtersCrown.jpeg

Exif Tool was run against the attachment to extract embedded metadata. Key findings are summarised below:

| Metadata Field | Value               |
|----------------|---------------------|
| File Name      | DaughtersCrown.jpeg |
| File Size      | 19 kB               |

| Metadata Field     | Value                             |
|--------------------|-----------------------------------|
| File Type          | JPEG                              |
| MIME Type          | image/jpeg                        |
| Image Dimensions   | 822 x 435 pixels                  |
| Resolution         | 120 x 120 DPI                     |
| JFIF Version       | 1.01                              |
| Encoding           | Baseline DCT, Huffman coding      |
| File Creation Date | 2021-01-25 16:41:00 +05:30        |
| Zone Identifier    | Exists (downloaded from internet) |

The Zone Identifier field confirms the file was downloaded from the internet. The file creation timestamp of 2021-01-25 aligns with the email send date, suggesting the attachment was prepared immediately before the campaign launch.

## 5.4 OSINT Investigation — emkei.cz

Open-source intelligence (OSINT) was conducted on the relay server emkei.cz. A Google search confirmed that emkei.cz is a publicly accessible Fake Mailer service — allowing anyone to send emails with any From address, no authentication required, operational since 2009.

| OSINT Field        | Finding                                                    |
|--------------------|------------------------------------------------------------|
| Service Name       | Emkei's Fake Mailer                                        |
| URL                | https://emkei.cz                                           |
| IP Address         | 93.99.104.210                                              |
| Operational Since  | 2009                                                       |
| Nature             | Anonymous fake email service — no sender authentication    |
| Threat Actor Use   | Abused for phishing, BEC, and social engineering campaigns |
| Header Fingerprint | Received: from localhost (emkei.cz [93.99.104.210])        |
| Threat Level       | <b>HIGH — known malicious infrastructure</b>               |

## 5.5 Hidden Content — Excel Attachment (Sheet3)

Analysis of an Excel attachment found within the email revealed a hidden sheet (Sheet3) using black background with white text — a concealment technique designed to hide content from casual inspection. The hidden cells contained Base64-encoded strings which upon decoding revealed:

**The Martian Colony, Beside Interplanetary Spaceport.**

This decoded value directly corresponds to the fictional location in the email body lore, confirming the attacker embedded encoded clues consistently across both the email and hidden attachment data.

| Finding            | Detail                                                              |
|--------------------|---------------------------------------------------------------------|
| Sheet Name         | Sheet3                                                              |
| Concealment Method | White text on black background — invisible to casual viewer         |
| Encoding           | Base64                                                              |
| Decoded Value      | <b>The Martian Colony, Beside Interplanetary Spaceport.</b>         |
| Analyst Action     | Select all cells, change font colour to black to reveal hidden text |
| Significance       | Attacker embedded encoded clues across multiple file types          |

## 6. Indicators of Compromise (IOCs)

| IOC Type      | Value                   | Context                                                        |
|---------------|-------------------------|----------------------------------------------------------------|
| IP Address    | 93.99.104.210           | <i>emkei.cz relay server — malicious sender infrastructure</i> |
| Domain        | emkei.cz                | <i>Anonymous fake email sender service abused by attacker</i>  |
| Domain        | microapple.com          | <i>Typosquat domain impersonating Apple Inc.</i>               |
| Email Address | billjobs@microapple.com | <i>Spoofed sender address</i>                                  |
| Email Address | negeja3921@pashter.com  | <i>Attacker-controlled Reply-To inbox</i>                      |
| Domain        | pashter.com             | <i>Disposable email service hosting attacker inbox</i>         |
| File          | DaughtersCrown.jpeg     | <i>Attachment — JPEG, 19kB, Zone ID exists</i>                 |
| File          | GoodJobMajor.pdf        | <i>Attachment — PDF, magic bytes 25 50 44 46</i>               |

## 7. Tools & Techniques Used

- Notepad++ — Raw .eml file inspection and header extraction
- CyberChef — Base64 decoding and hex conversion of encoded email body
- filesig.search.org — File signature (magic bytes) lookup and verification
- ExifTool — Metadata extraction from JPEG and PDF attachments
- Manual Header Analysis — Field-by-field inspection of SMTP routing, authentication, and spoofing indicators

## 8. Recommendations

### 8.1 Immediate Actions

- Block IP 93.99.104.210 and domain emkei.cz at the email gateway and firewall
- Block sender domain microapple.com as a known typosquat
- Add negeja3921@pashter.com and billjobs@microapple.com to threat intelligence blocklists
- Submit IOCs to VirusTotal and internal SIEM for retroactive hunting

### 8.2 Detection Engineering

- Create SIEM rule to alert on emails where Reply-To differs from From address
- Create rule to flag emails originating from emkei.cz or similar anonymous relay services
- Alert on SPF FAIL + DKIM absent + no DMARC — triple authentication failure pattern
- Use Message-ID containing 'localhost' as an indicator of fake mailer usage

### 8.3 User Awareness

- Train users to verify Reply-To address before responding to any email
- Educate staff on typosquatting — microapple.com vs apple.com distinction
- Implement a mail client banner warning for external emails with mismatched Reply-To

## 9. Final Verdict

### **VERDICT: MALICIOUS — PHISHING EMAIL**

The email "A Hope to CoCanDa" is a confirmed phishing email. The following factors collectively establish this verdict with high confidence:

- SPF authentication FAIL — sending IP not authorised for the claimed domain
- Sender relayed through emkei.cz — a known anonymous fake email service
- Reply-To mismatch — attacker redirecting victim replies to a controlled inbox
- Typosquat sender domain (microapple.com) impersonating Apple Inc.
- Base64-encoded body with social engineering lure content
- Attachments delivered with zone identifier confirming internet origin

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